DEVOPS TOOLS ASSIGNMENT – 1

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TITLE:

"HOSPITAL MANAGEMENT SYSTEM"

FULL TITLE: **Hospital Management System (full-stack)** — a role-based hospital operations platform built with Git/GitHub for version control, and Docker & Jenkins for containerized deployment automation.

DESCRIPTION:

The Hospital Management System is a full-stack web application built to manage day-to-day hospital operations across five distinct roles — Admin, Doctor, Nurse, Receptionist and Patient. Each role signs in to its own dashboard and sees only the data and actions relevant to it: admins oversee users, doctors and departments hospital-wide; doctors manage their own patients and appointments; nurses assist with patient care; receptionists handle registration and scheduling; and patients book appointments and view their own records.

The application was developed as a demonstration of how a real, data-backed full-stack application can be integrated with modern DevOps practices. The project uses Git and GitHub for source-code management, Node.js/Express for the backend REST API, React for the frontend, and MySQL via the Sequelize ORM as the database — with Docker and Jenkins included in the repository to automate containerized deployment.

The system covers patient records, appointment scheduling with double-booking prevention, prescriptions, medical records and billing. Unlike a JSON-file backed demo, data is persisted in a real MySQL 8 database across nine migrated tables with seeded demo data, and every endpoint is secured with JWT authentication and per-role authorization middleware.

The main objective of the project is not just a UI demo, but to demonstrate the complete lifecycle of a full-stack application from development and version control, through automated testing, to containerized deployment.

TOOLS USED:

TOOLS	PURPOSE
React 18 + Vite	Frontend SPA — component-based UI, dev server
React Router	Client-side routing between role dashboards
Tailwind CSS	Styling, layout and responsive design
Axios	Frontend → REST API communication
Node.js	Backend programming
Express	REST API and application server

Sequelize	MySQL schema modelling, migrations and seeders
MySQL 8	Relational application data storage
JWT / bcryptjs	Authentication and password hashing
Jest / Supertest	Automated API testing
git	Version control
github	Remote source code repository
docker	Application containerization
jenkins	CI/CD automation

PROJECT STRUCTURE:

The project is organized into separate frontend, backend, database and deployment components.

The `backend` directory contains the Express application — controllers, routes, models, middleware and services. The `frontend` directory contains the React user interface. The `database` directory contains the Sequelize migrations and seeders. The `Dockerfile`s (one per service) define the container images, while the `Jenkinsfile` defines the CI/CD pipeline.

BACKEND IMPLEMENTATION:

The backend was implemented using **Node.js and the Express framework**.

The Express application provides REST API endpoints for accessing hospital data. The major endpoints implemented are:

```
GET    /api/health
POST   /api/auth/login
GET    /api/auth/me
GET / POST / PUT / DELETE    /api/patients
GET / POST / PUT / DELETE    /api/doctors
GET / POST / PUT / DELETE    /api/appointments
GET / POST / PUT / DELETE    /api/medical-records
GET / POST / PUT / DELETE    /api/prescriptions
GET / POST / PUT / DELETE    /api/bills
GET    /api/dashboard
```

The `/api/dashboard` endpoint provides aggregated, role-aware statistics such as total patients, doctors, nurses, appointments (pending / completed / cancelled), prescriptions, medical records and billing totals — this is what powers every role's dashboard screen.

The `/api/appointments` endpoints prevent double-booking a doctor for the same slot, and support confirm / reject / complete state transitions restricted to doctors and admins.

The backend also implements appropriate HTTP responses for invalid IDs. For example, requesting a patient that does not exist produces an HTTP 404 Not Found response rather than causing the application to crash.

FRONTEND IMPLEMENTATION:

The frontend was developed using React 18, Vite and Tailwind CSS.

Each role lands on its own dashboard after login:

- Admin dashboard
- Doctor dashboard
- Nurse dashboard
- Receptionist dashboard
- Patient dashboard
- Role-based route protection
- Shared appointments, bills, doctors, patients, prescriptions and records pages
- Toast notifications

When a dashboard is loaded, the React app requests data from the Express REST API via Axios and renders it into live counters and tables.

When a user books, confirms or updates an appointment, the frontend calls the corresponding API endpoint and refreshes the affected views.

This creates a clear separation between the frontend presentation layer and the backend data / API layer.

TESTING AND AUTOMATED TESTING:

The application was executed locally and accessed through:

`http://localhost:5173`

The dashboards, REST APIs, authentication and role-based access were all verified.

A dedicated test suite was created using **Jest** and **Supertest**.

The test suite contains 112 automated test cases across 7 suites. The tests verify:

1. Registration and login.
2. JWT-protected routes and rejection of missing/malformed tokens.
3. Role-based authorization (e.g. a patient is blocked from the admin-only users endpoint).
4. Patient, doctor, nurse and department CRUD, search and pagination.
5. Appointment booking, confirm/reject/complete transitions, and double-booking prevention.
6. Handling of invalid IDs with proper 404 responses.
7. Dashboard summary counters for every role.
8. Health check liveness and readiness.

The tests were executed using:

```
$ npm test
> cross-env NODE_ENV=test jest --runInBand --forceExit
```

The final test execution produced:

```
PASS  tests/health.test.js
PASS  tests/doctor.test.js
PASS  tests/patient.test.js
PASS  tests/auth.test.js
PASS  tests/bill.test.js
```

```
PASS  tests/prescription.test.js
PASS  tests/appointment.test.js

Test Suites: 7 passed, 7 total
Tests:       112 passed, 112 total
Time:        13.374 s
```

This automated test suite forms the primary Continuous Integration verification step in the Jenkins pipeline (see below).

GIT AND GITHUB INTEGRATION:

The project is version-controlled with Git. A private GitHub repository was created and the initial commit pushed to it, so that Jenkins can retrieve the latest version of the project automatically. GitHub also provides a central location for maintaining the source code and the Jenkinsfile.

```
$ git remote -v
origin  https://github.com/Varshith8999/hospital-management-system.git (fetch)
origin  https://github.com/Varshith8999/hospital-management-system.git (push)

$ git log --oneline
5b0d67e Initial commit: Hospital Management System

$ git status
On branch main
Your branch is up to date with 'origin/main'.
```

Repository: **github.com/Varshith8999/hospital-management-system** (private). `.env`, `node_modules/` and all credentials are excluded via `.gitignore`; only `.env.example` (a template with no real secrets) is tracked.

DOCKER CONTAINERIZATION:

> **Note:** Docker and Jenkins are fully *defined* in this repository and ready to run, but were not executed for this report — the app below was run and verified directly with Node.js and MySQL instead. This section documents the deployment design as written, not a verified container run.

Docker was included to package the application into portable containers.

`docker-compose.yml` defines three services:

- `mysql` — MySQL 8, persistent volume, healthcheck
- `backend` — built from `backend/Dockerfile`, waits for MySQL's healthcheck, runs migrations + seeders on start
- `frontend` — built from `frontend/Dockerfile`, served by nginx, proxies `/api` to the backend

The backend `Dockerfile` performs the following operations:

1. Selects the Node.js 18 slim base image.
2. Sets the working directory to `/app`.
3. Copies `package.json` and installs production dependencies.
4. Copies the project files, including the shared `database/` migrations and seeders.
5. Exposes port 5000.
6. Runs migrations/seeders (when enabled) and starts the Express application.

Intended to be started with:

```
$ cp .env.example .env   # then set real credentials
$ docker compose up --build
```

JENKINS CONFIGURATION:

The repository includes a declarative `Jenkinsfile`, designed to connect to the GitHub repository above and run with access to Git, Node.js and Docker. It defines four stages:

`Checkout → Test → Docker Build → Deploy`

CONTINUOUS INTEGRATION AND DEPLOYMENT:

The Checkout stage retrieves the latest source code from GitHub.

The Test stage is designed to execute the 112-test Jest suite shown above. If a test fails, the pipeline stops.

The Docker Build stage builds the backend and frontend images after successful testing.

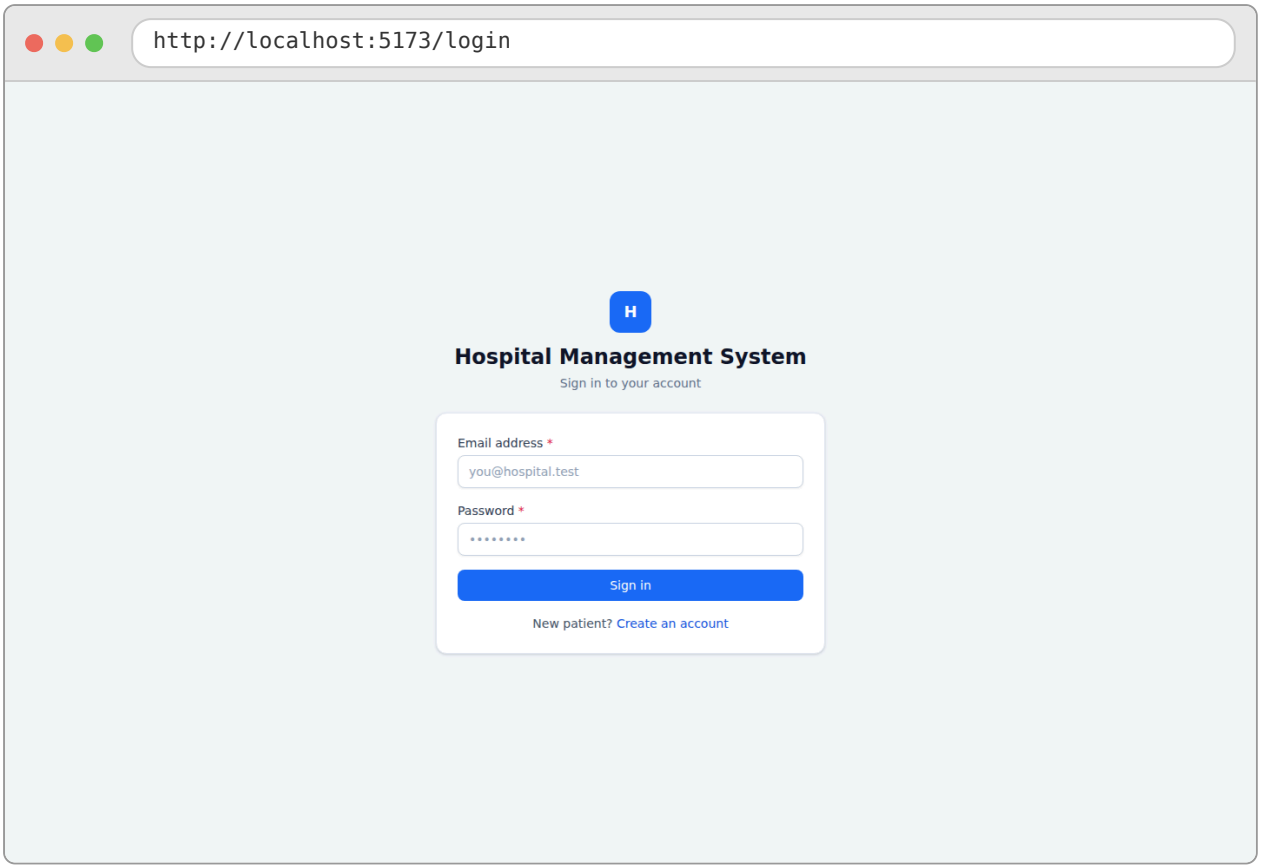
The Deploy stage brings the stack up via Docker Compose and is designed to verify the containers are running.

LOCAL VERIFICATION:

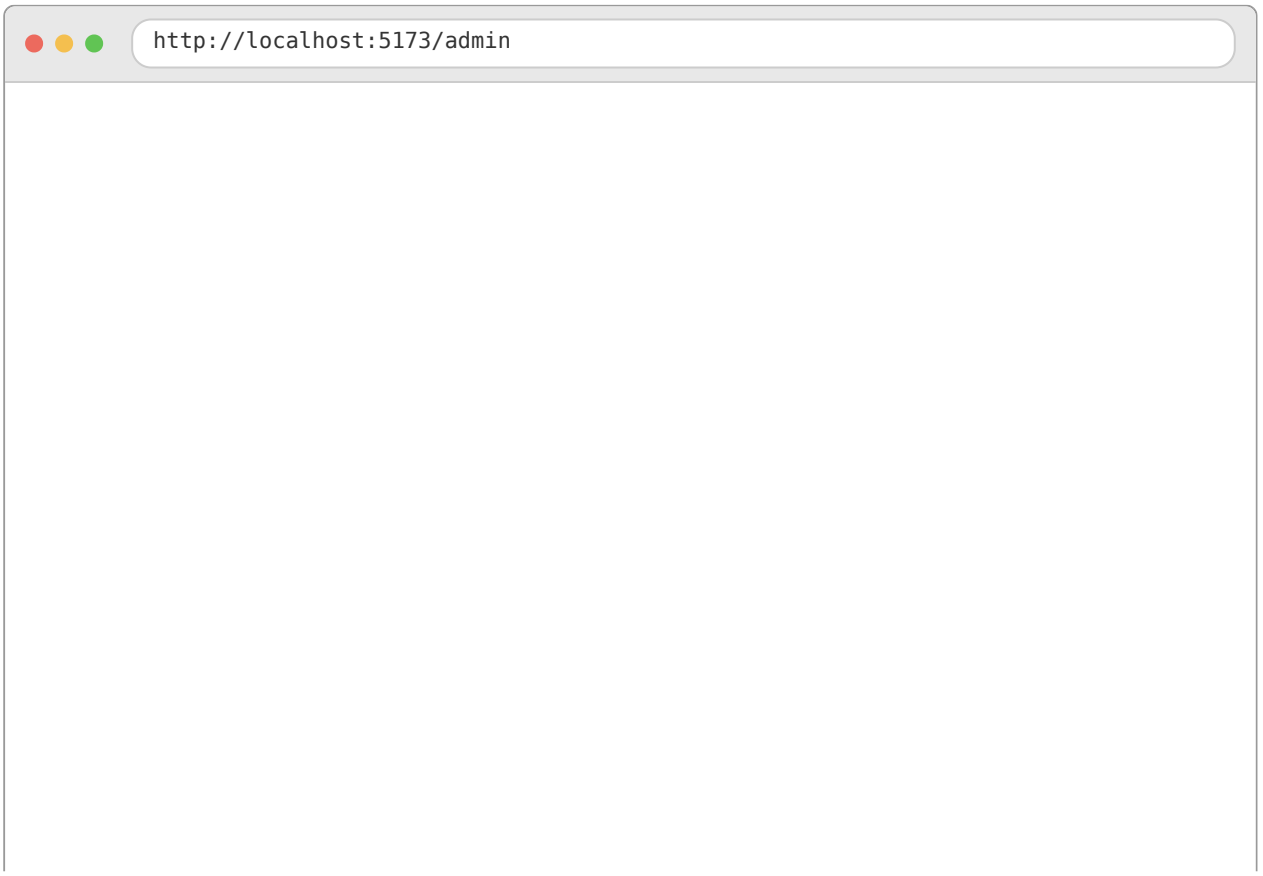
In place of the containerized pipeline above, the application was actually run and verified directly: Node.js 18 and MySQL 8 inside a local Linux (WSL2) environment, migrations and seeders applied against a real MySQL database, then the backend and frontend started and exercised through a real browser session.

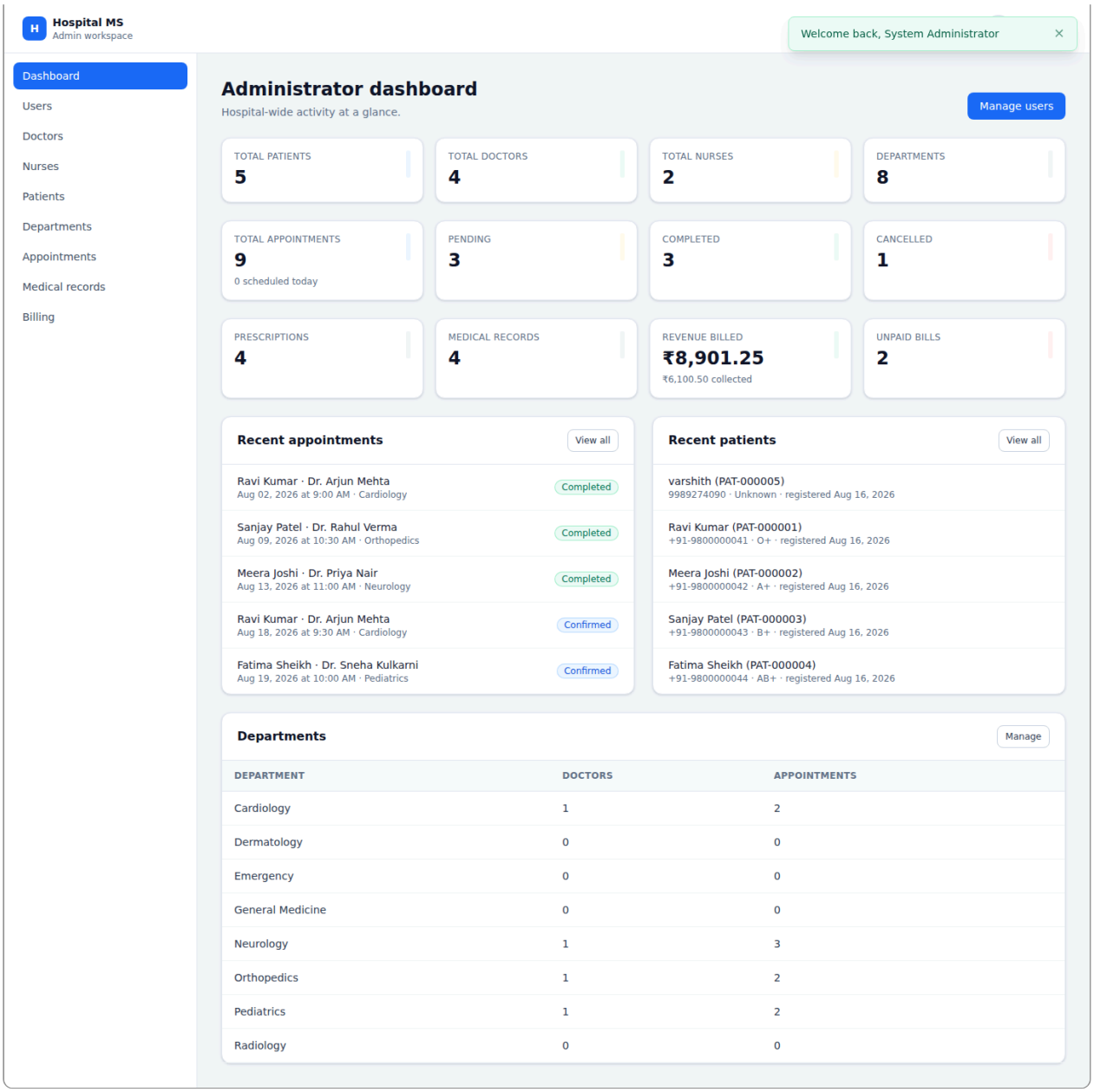
```
$ curl http://localhost:5000/api/health
{"status":"ok","service":"hospital-management-backend","dependencies":
{"database":"up"}}

$ curl -X POST http://localhost:5000/api/auth/login -d
'{"email":"admin@hospital.test","password":"Password@123"}'
{"success":true,"message":"Login successful", "token":"eyJhbGciOi..."}
```

RESULT:



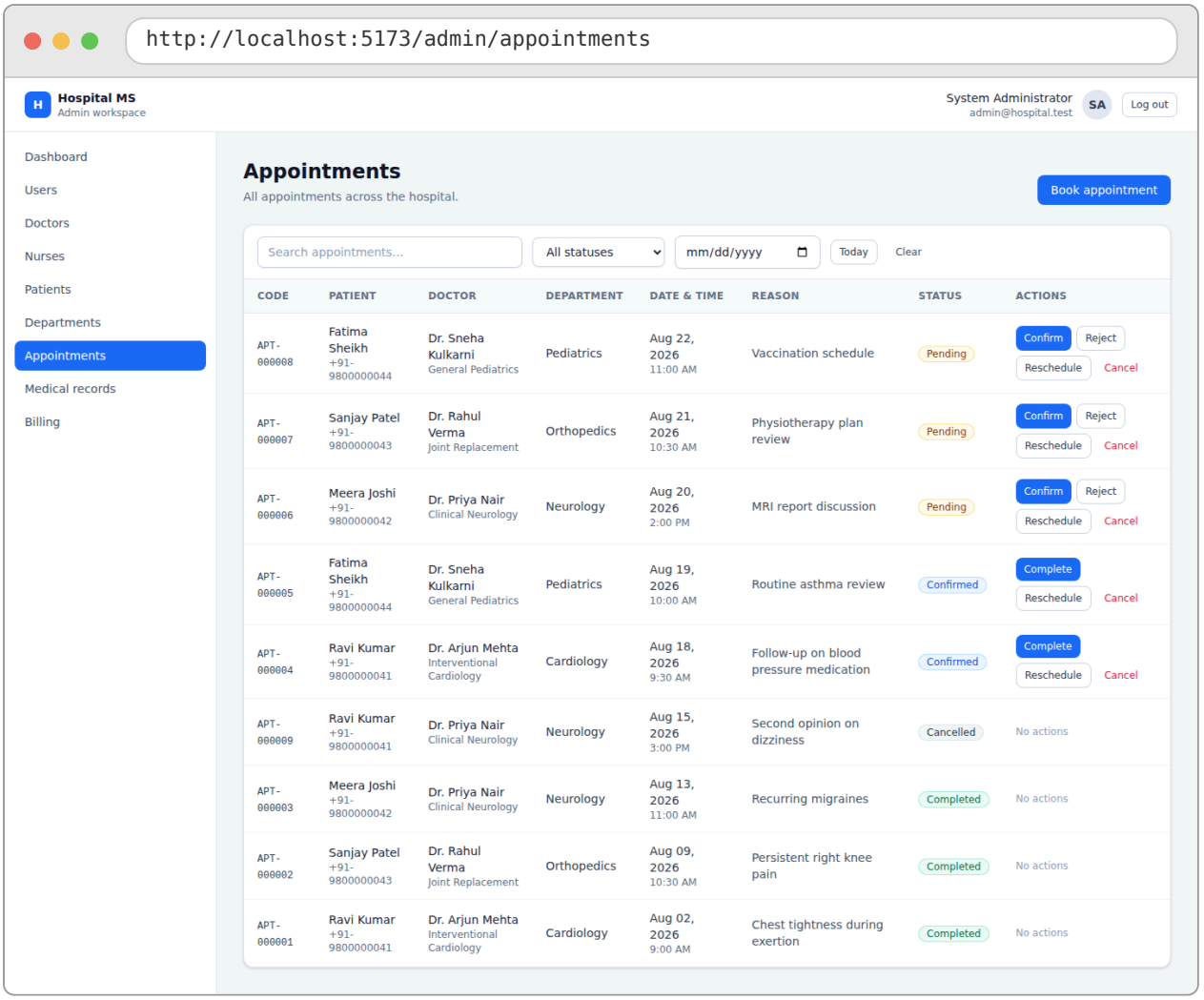
Login screen.





Administrator dashboard after login — live counts of patients, doctors, nurses, departments, appointments, prescriptions, medical records and billing, pulled from the real MySQL database via `/api/dashboard`.

TESTING:



Appointments management — real seeded appointment data with confirm / reject / complete / reschedule / cancel actions, exercised as the Admin role.

CONCLUSION:

The Hospital Management System demonstrates the development of a production-shaped full-stack web application integrated with modern DevOps practices.

The project combines a Node.js/Express REST backend with a React frontend to provide role-based dashboards for hospital operations. MySQL, managed through versioned Sequelize migrations, was used to keep the data model realistic and relational rather than a flat-file demo.

Git and GitHub were used for source-code management and version control, while Jest and Supertest provided 112 automated API tests, all passing. Docker and Jenkins are included in the repository to containerize the application and automate deployment, ready to be executed even though this run verified the application directly via a native Node.js/MySQL setup.

The project demonstrates that CI/CD practices can be applied effectively to a real, data-backed application. Natural next steps include actually executing the Docker Compose stack and the Jenkins pipeline end-to-end, adding HTTPS and production-grade secrets management, and extending the test suite to cover billing and prescription workflows under load.